

HSINCHU, Taiwan

WMO: 467560

Lat: 24.818N

Lon: 120.939E

Elev: 8

StdP: 101.23

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.1	10.2	5.0	5.4	10.3	6.8	6.1	11.9	13.4	14.9	12.1	13.9	6.5	40	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	33.1	27.9	32.6	27.7	32.1	27.5	28.9	32.0	28.3	31.5	27.9	31.3	6.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.0	24.2	31.5	27.2	23.0	30.9	27.0	22.7	30.8	94.3	32.0	91.5	32.0	89.9	31.3	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.5	12.0	10.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			5.4	30.0	1.6	0.9	4.3	30.7	3.3	31.2	2.4	31.8	1.2	32.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	15.8	15.7	17.4	21.6	24.8	27.4	29.0	29.0	27.3	24.3	21.1	17.7		
	DBStd	5.37	2.38	2.74	3.13	3.15	2.41	2.00	1.03	1.11	1.85	1.98	2.62	2.39		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	276	84	81	54	10	1	0	0	0	0	0	6	40		
	CDD10.0	4613	179	161	231	347	459	522	589	589	518	444	334	239		
	CDD18.3	1846	4	8	27	107	201	272	331	331	268	186	91	21		
	CDH23.3	17381	5	21	82	517	1539	2942	4080	4067	2714	1084	300	29		
	CDH26.7	6144	0	1	8	62	366	1061	1743	1708	930	224	39	1		
(14) Wind	WSAvg	4.8	5.6	5.4	4.7	4.0	3.3	4.2	4.0	4.0	4.2	6.5	6.7	5.9		
(15) Precipitation	PrecAvg	1623	65	131	151	164	206	250	125	171	198	62	51	49		
	PrecMax	2344	154	364	275	503	677	579	344	445	1192	252	222	137		
	PrecMin	779	13	14	77	18	16	2	0	0	0	0	0	0		
	PrecStd	461	33	98	45	120	134	155	98	145	279	70	60	39		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.1	25.8	27.9	29.2	31.9	33.0	33.8	34.2	34.2	31.2	29.7	25.8		
		MCWB	19.7	22.0	23.7	25.3	27.3	27.8	28.1	28.2	27.7	26.0	24.3	21.1		
	2%	DB	22.2	23.4	25.4	28.2	30.8	32.1	33.0	33.2	32.2	29.9	27.8	24.0		
		MCWB	18.7	20.1	22.6	24.7	26.5	27.5	27.9	28.1	27.2	25.6	23.4	20.2		
	5%	DB	20.9	21.8	23.8	27.2	29.8	31.3	32.2	32.8	31.8	28.9	26.2	22.9		
		MCWB	17.8	18.7	21.5	24.0	25.8	27.2	27.6	27.9	27.0	24.5	22.5	19.3		
	10%	DB	19.2	19.9	22.1	26.1	28.9	31.0	32.0	32.0	30.9	28.0	25.0	21.7		
		MCWB	16.9	17.4	19.9	23.3	25.4	27.1	27.5	27.6	26.7	24.0	21.8	18.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	22.3	24.3	26.1	28.1	29.0	29.2	30.0	28.9	27.0	25.1	22.4		
		MCDB	23.1	25.0	27.0	28.3	31.2	31.7	32.1	32.6	32.0	30.2	27.8	24.6		
	2%	WB	19.2	20.5	22.8	25.3	26.7	28.2	28.7	29.0	28.0	26.1	23.9	20.9		
		MCDB	21.7	22.9	25.1	27.6	29.6	31.1	31.8	32.2	31.5	28.9	27.1	23.0		
	5%	WB	18.3	19.0	21.6	24.4	26.2	27.7	28.2	28.3	27.4	25.2	23.0	20.0		
		MCDB	20.3	21.2	23.5	26.7	28.9	30.7	31.5	31.6	30.8	28.1	25.4	22.1		
	10%	WB	17.4	17.6	20.3	23.7	25.6	27.1	27.8	27.9	26.9	24.5	22.2	19.0		
		MCDB	19.0	19.6	21.9	25.9	28.2	30.2	31.2	31.2	30.1	27.1	24.5	21.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	6.1	5.8	6.0	6.0	6.1	6.3	6.2	6.5	6.5	6.3	6.7		
		MCDBR	8.6	9.2	8.5	7.0	6.9	6.6	6.6	7.1	7.5	7.7	8.1	8.6		
	5% WB	MCWBR	6.1	6.6	5.9	4.9	4.5	4.2	4.0	4.3	4.4	4.8	5.1	5.8		
		MCWBR	7.3	8.9	8.0	6.4	6.4	6.5	6.4	6.5	6.8	6.9	7.1	7.6		
(40) Clear-Sky Solar Irradiance	taub		0.540	0.587	0.699	0.703	0.599	0.555	0.551	0.572	0.554	0.539	0.530	0.536		
		taud	1.888	1.806	1.604	1.645	1.912	2.049	2.055	1.988	2.016	1.999	1.996	1.914		
	Ebn at Noon		701	701	648	661	732	760	762	746	749	734	707	684		
		Edh at Noon	177	205	264	258	197	171	169	181	172	167	158	167		
(44) All-Sky Solar Radiation	RadAvg		2.45	2.74	3.34	3.74	4.35	4.97	6.00	5.44	4.75	3.92	2.81	2.43		
	RadStd		0.44	0.49	0.42	0.54	0.74	0.58	0.36	0.50	0.50	0.43	0.33	0.37		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		+0.25	N/A	N/A	+0.33	N/A	N/A	N/A	N/A	+123	+97

Nomenclature: See separate page